

Joseph Busbin

<https://www.linkedin.com/in/joseph-busbin/>

Education

William & Mary, Williamsburg, VA

Bachelors of Science

May 2027

Majors: Computational and Applied Mathematics and Statistic; Biology

Select Coursework: Data structures, Algorithms, Software Development, Multivariable Calculus, Linear Algebra, Foundations of Math, Ordinary Differential Equations, Mathematical Modeling in Biology, Probability, Molecular Cell Biology, Microbiology, Animal Behavior, Organic Chemistry, General Chemistry, Introduction to Psychology, Introductory Physics

Skills

- Technology: Python, Java, C++ , MatLab, Git/Github, SQL; proficient in Software Development practices, Excel, Microsoft Suite
- Math: Probabilistic methods, constructing and interpreting mathematical models, proof reading and writing, Multivariate Calculus, Ordinary Differential Equations
- Biology Lab techniques: streaking for isolation, culturing, dying, genetic screening, antibiotic resistance, transduction, phage mediated gene selection, observational study and experimental behavioral study
- Chemistry Lab techniques : Purification and analysis of organic and inorganic compounds, Spectroscopy (H, C13), gas chromatography, thin layer chromatography, extractions, titrations, compound analysis

Research Experience

- Literature review and experiment proposal on nanoparticle delivery vehicles to the nucleus for gene repair
 - Creation of an academic paper in biology with mock implementation of methods and results which deepened my understanding of experimental design, while applying knowledge of cell biology and protein function.

- Refined abilities of literature search, research writing capability, and grasp of ethical research practice.
- Mitochondria population dynamic modeling
 - Class project: Reconstructing a research paper and modifications
 - Selected topic, conducted literature review and understood the mathematical models used.
 - Reconstructed simulation of mathematical model in MATLAB and solved both numerically as well as symbolically.
 - Extended the model to simulate novel, and more realistic behaviors of the biological system

Project Experience

- Building data structures and Huffman tree encoding in Python
 - A Huffman tree used to find optimal custom encodings for signal transmission. Building it in Python entailed understanding and engineering the tree data structures that allowed for consistently minimized representations for the components of a message.
- Puzzle game development in Java
 - Implemented multiple puzzles creating back-end logic to interact with a display GUI according to the design of popular games like Wordle and 2048.
- 2025 Tribe Capture The Flag (CTF)
 - This 2 day event was a journey into the world of cyber security. It involved attempting a series of combined challenges related to coding, forensics, cryptography, hardware and more.

Leadership Experience

- Volunteer Advisory Council *Atrium Health Wake Forest Baptist Hospital*, Winston Salem, NC July - August 2021
 - Leadership of roughly 50 other volunteers as part of a team of mentors who guided, trained and assisted the volunteer department in relation to student

volunteer operations.

- Promoted and displayed a culture of service and mentality of positivity though the group while respecting the often difficult aspects of patients/care.

Volunteer Experience

- *Atrium Health Wake Forest Baptist Hospital*, Winston Salem, NC – Volunteer June - August 2019
 - Shadowed positions throughout hospital departments from medical staff to administration learning role specific functions and gaining a wide understanding of hospital and healthcare operations
 - Learned to give direct interpersonal patient care

Work Experience

- Williamsburg Occasions LLC – Bed and Breakfast Staff; 06/26-Present, Williamsburg VA
 - Creating a delightful and personalized experience for wonderful guests. Serving a full breakfast while respecting personal preference, dietary restrictions and health or allergy concerns every morning.
 - Providing personalized customer service and practicing clear and effective communication.
 - An experience in small business logistics, highly independent work, and problem solving.

Activities

- Pre-health Club
 - Healthcare related club; focused on understanding the medical field, careers, participating as a student, and becoming a medical professional.
- Computational and Applied Mathematics and Statistics (CAMS) club
 - Covers topics related to the interdisciplinary nature of CAMS, its research direction, applications, and careers.
- Association for Computation Machinery (ACM) club
 - Student chapter of ACM at William and Mary. Computer science related workshops, career development, and the coordination of special events.
- Robotic club

- A club dedicated to learning the building of robots, both software and hardware.
- Finance Academy
 - Student led program which provides resources, career development, and hosts recruiters from the finance industry. Primarily focuses on banking, investment, and corporate finance.
- Braun Financial Literacy Program
 - A program with a multi-year curriculum aimed at the general student body to help to prepare students for managing personal finance.

Awards

- Deans list, William and Mary: 2024, 2025, 2026

Last updated: July 2026